

WGCNA Analysis of mRNA Expression Data: Pipeline Structure, Results, and Signed vs. Unsigned Network Comparison

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August 2026

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1 Pipeline Structure

The full pipeline is implemented in a single R script (`mRNA.WGCNA.R`) built around the `WGCNA`, `DESeq2`, and `clusterProfiler` packages. It proceeds linearly from raw counts to modules, module–trait correlations, GO enrichment, and hub-gene extraction. This section summarises the logic of each stage, with particular attention to the four points requested: the size-factor export, the MAD histogram and sample-clustering dendrogram, the rationale for MAD over plain variance, the module–trait relationship logic, and the global parameters fixed at the top of the script.

1.1 Global Parameters

The script first fixes a small set of global parameters that control the entire run:

- `MIN_COUNT = 10`, `MIN_FRAC = 0.50`: the low-expression filter. A gene is retained only if it has at least 10 reads in at least 50% of the 963 samples. This is included to remove genes with negligible expression, not to do any variance-based curation, which will be performed by MAD further.
- `N_GENES_WGCNA = NA`: controls whether the gene set is truncated to the top- N most variable (by MAD) genes before WGCNA. Setting this to `NA` means *every* gene that survives the expression filter is carried into WGCNA; no additional hard cutoff is imposed.
- `MAX_BLOCK_SIZE = 12000`: raised well above the WGCNA default of 5000 so that all 7220 genes are processed as a single block.
- `N_HUBS = 20`: the number of top-kME genes exported per module for the hub-gene tables.

1.2 Loading, Low-Expression Filtering, and Size-Factor Export

The raw file (`mRNA_filter1.xlsx`) is read as a 963×8693 sample-by-column table, where one column is the PAM50 label (with “TNBC” relabelled to “Basal” to match standard nomenclature) and the remaining 8692 columns are raw counts. The matrix is transposed to the conventional genes-by-samples orientation for DESeq2, and the low-expression filter described above is applied, dropping the gene set from 8692 to 7220.

DESeq2 is then used, but in a **restricted role**: the design formula is ~ 1 (an intercept-only model), so DESeq2 is *not* being used here to test for differential expression. It is used purely for normalisation, specifically to run `estimateSizeFactors()` (median-of-ratios normalisation, which corrects for differences in sequencing depth across the 963 samples) and `estimateDispersions()` (the mean–variance trend that the variance-stabilising transform needs). The resulting size factors are written out explicitly:

```
saveRDS(sizeFactors(dds_mrna),
        file.path(OUT_DIR, "mrna_size_factors.rds"))
```

This export exists so that the lncRNA pipeline can reuse the *same* size factors rather than re-estimating its own. *The lncRNA count submatrix is far sparser (lower per-gene counts, more zeros) than the mRNA matrix*, so DESeq2’s size-factor estimator is less reliable if run on the lncRNA data in isolation (library depth is a property of the sequencing run, not of the transcript biotype), so it is more robust to estimate it once from the denser mRNA counts and carry it across. After normalisation, `varianceStabilizingTransformation()` (blind to the subtype labels) produces the VST matrix that all downstream steps operate on.

1.3 MAD-Based Feature Selection and Why MAD Over Variance

Why MAD Over Standard Variance for Feature Selection:

- **Outlier Sensitivity:** MAD is defined as $\text{median}(|x_i - \text{median}(x)|)$, whereas variance relies on squared deviations from the mean ($|x_i - \bar{x}|^2$). Squaring heavily penalizes large deviations, allowing a small number of extreme samples (e.g., mis-normalization, batch artifacts, or extreme biological outliers) to artificially inflate a gene's variance score.
- **Cohort-Wide Consistency:** MAD selects genes based on consistent, reproducible expression variability across the whole dataset rather than variance driven by extreme values in a few samples. *It is also recommended by standard WGCNA tutorials.*

In this run, `N_GENES_WGCNA` was set to `NA`, so no hard MAD cutoff was actually applied. The histogram in Figure 1 was generated as a diagnostic (and the code contains a conditional to draw a vertical cutoff line on it, for use in future runs where a top- N truncation is desired), but for this analysis all 7220 expression-filtered genes were retained and passed straight into WGCNA.

1.4 Sample Clustering (Outlier Screening)

Before network construction, samples are hierarchically clustered (average linkage on Euclidean distance across all 7220 genes) purely as a visual outlier check:

```
sampleTree_mrna <- hclust(dist(datExpr_mrna), method = "average")
plot(sampleTree_mrna, ...)
CUT_HEIGHT_MRNA <- NA # set to a numeric height to actually cut
```

The cut height, for this run, is left as `NA` in the script. The dendrogram is plotted and inspected first, and only if a genuinely isolated cluster of arrays is visible at some height would that height be filled in and a `cutreeStatic()` call used to drop those samples. No such height was set in this run, so all 963 samples that passed `goodSamplesGenes()` were retained.

1.5 Module–Trait Relationship Logic

PAM50 subtype is a four-level categorical variable, so it cannot be correlated with a module eigengene directly. The script converts it into a one-hot trait matrix:

This produces four binary indicator columns (Basal, HER2, LumA, LumB), each a 0/1 column across the 963 samples.

For each module, the eigengene (first principal component of that module's expression sub-matrix, one value per sample) is correlated against each of the four indicator columns using Pearson correlation, and the corresponding p -value is obtained from `corPvalueStudent()`, which converts a correlation coefficient and sample size into a Student's t -based p -value under the null hypothesis of no association. *This is repeated once for the signed network's eigengenes and once for the unsigned network's, producing the two module–trait heatmaps (Figures 8 and 9) and their corresponding CSV tables.*

1.6 Network Construction, GO Enrichment, and Hub Genes (brief)

Soft-thresholding power is chosen automatically via `pickSoftThreshold()` across candidate powers $\{1, \dots, 10, 12, 14, 16, 18, 20\}$, separately for signed and unsigned adjacency, taking the lowest power at which the scale-free topology fit R^2 first exceeds 0.90 (falling back to a manual default if no power reaches that target). `blockwiseModules()` was then run twice for each network type: the first run (`deepSplit=2, minModuleSize=30`) required some hyperparameter tuning, (`deepSplit=1, minModuleSize=50, same mergeCutHeight=0.25`). The tuned version

actually carried forward into every downstream analysis (module colours, eigengenes, module-trait correlations, GO enrichment, hub genes).

2 Results

2.1 Cohort and Preprocessing Summary

The starting cohort comprised 963 tumour samples with PAM50 subtype calls distributed as shown in Table 1. Class imbalance (LumA making up more than half the cohort, HER2 well under 10%) is observed.

Table 1: PAM50 subtype composition of the 963-sample cohort.

	Basal	HER2	LumA	LumB
Samples	170	75	528	190

Of the 8692 raw genes, 7220 (83.1%) passed the low-expression filter (mean detection ≥ 10 reads in $\geq 50\%$ of samples). Library sizes ranged from 7.3M to 45.7M reads (median 22.7M), and DESeq2 size factors ranged from 0.374 to 2.048 (median 1.029), indicating a roughly $5.5\times$ spread in effective sequencing depth across the cohort that the size-factor normalisation and subsequent VST are correcting for.

2.2 MAD Distribution

Figure 1 shows the distribution of gene-level MAD on the VST scale across the 7220 expression-filtered genes. The distribution is strongly right-skewed: the bulk of genes sit in a fairly narrow band of low-to-moderate MAD (roughly 0.5–1.5), with a long thin tail extending out past MAD = 4–6 for a small number of highly variable genes. Because `N_GENES_WGCNA` was left at NA, no cut was drawn on this distribution for the present run.

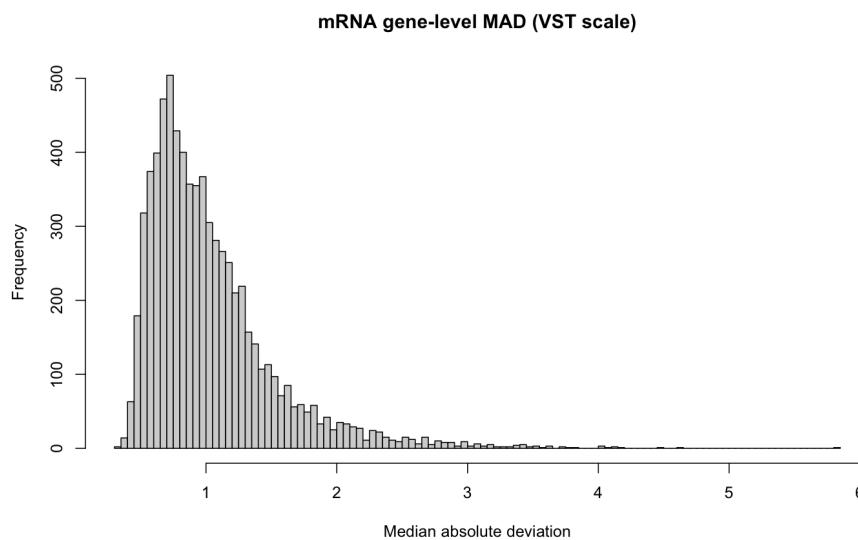


Figure 1: Distribution of gene-level MAD (VST scale) across the 7220 expression-filtered mRNA genes.

2.3 Sample Clustering

Figure 2 shows the average-linkage sample dendrogram used for outlier screening. There is no single sample or small sample cluster sitting at a dramatically greater height than the rest of the tree in a way that would indicate a technical failure; the tree instead shows the expected

fine-grained substructure of a large, biologically heterogeneous tumour cohort. On this basis no cut height was applied and all 963 samples were retained.

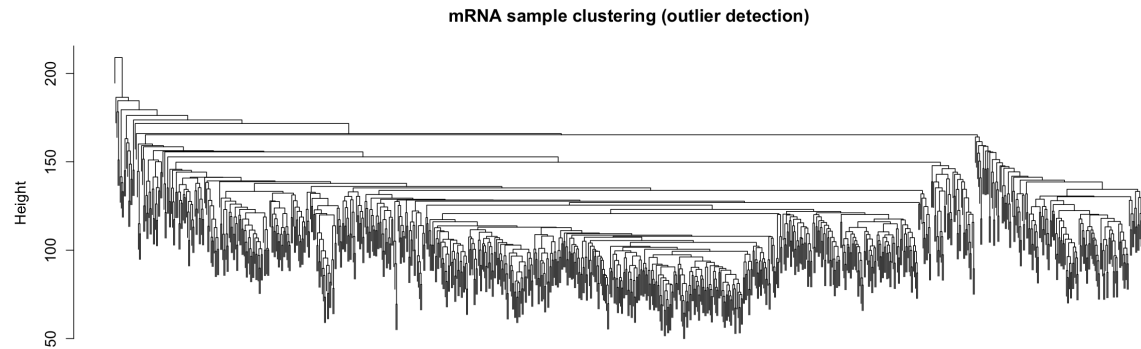


Figure 2: Average-linkage hierarchical clustering of all 963 samples (outlier screening dendrogram). No cut was applied.

2.4 Soft-Threshold Selection

Figure 3 shows the scale-free topology fit (R^2 , top row) and mean connectivity (bottom row) curves for both network types across the candidate power sequence.

Table 2: Selected soft-thresholding power and fit statistics for each network type.

Network type	Chosen power	Scale-free R^2 at chosen power	Mean connectivity
Unsigned	7	0.923	4.50
Signed	14	0.917	8.25

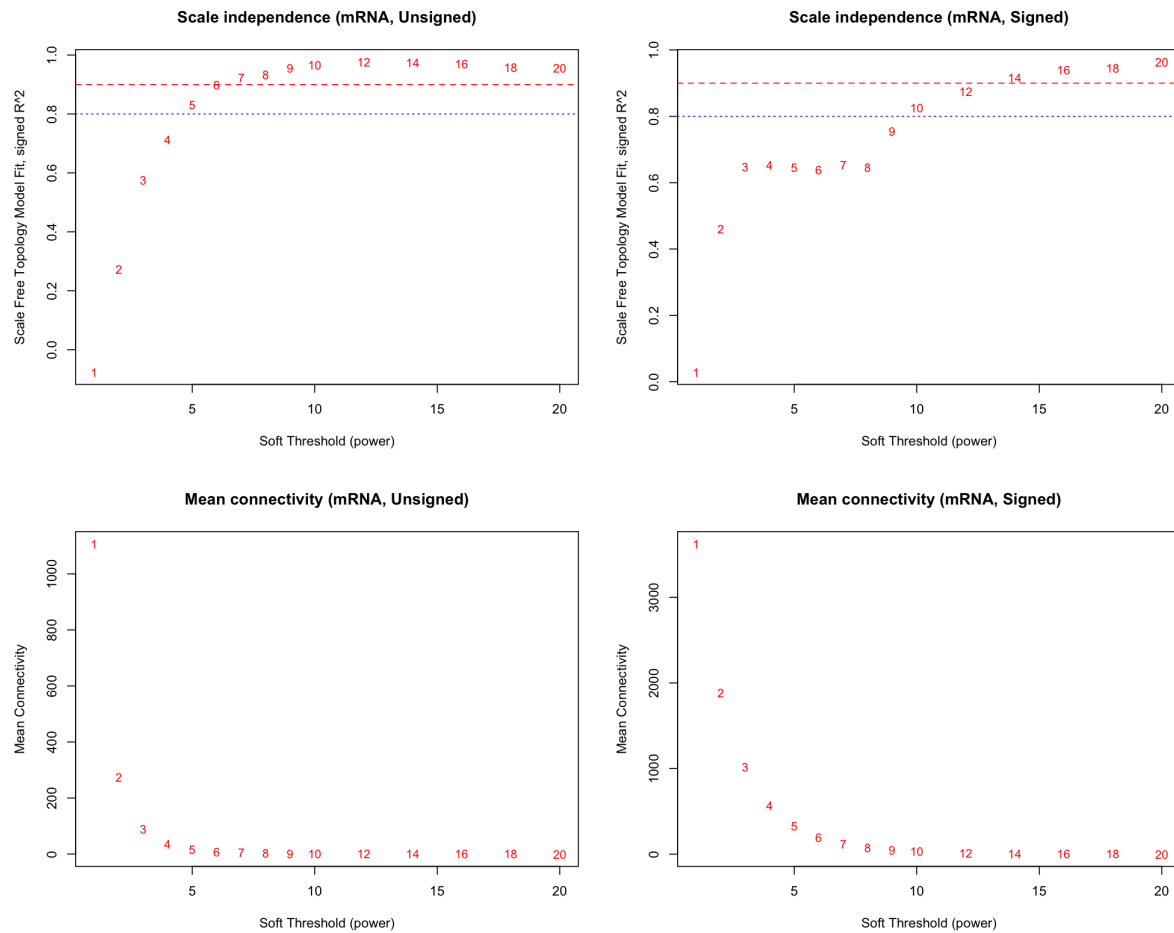


Figure 3: Scale-free topology fit and mean connectivity as a function of soft-thresholding power, for the unsigned (left column) and signed (right column) networks.

2.5 Network Construction and Module Detection

Table 3 reports the final ("tuned", `deepSplit=1`, `minModuleSize=50`) module sizes for both network types. The signed network resolves into 9 non-grey modules covering 4050 of 7220 genes (56.1%), while the unsigned network resolves into 7 non-grey modules covering only 2644 of 7220 genes (36.6%). *The unsigned grey module is nearly 1.5 $\times$ larger in gene count than the signed grey module (4576 vs. 3170 genes).*

Table 3: Final tuned module sizes (gene counts) for the signed and unsigned mRNA networks. Grey denotes unassigned genes.

Module	Signed	Unsigned
black	262	66
blue	715	637
brown	619	516
green	503	300
magenta	65	—
pink	85	—
red	472	72
turquoise	785	639
yellow	544	414
grey (unassigned)	3170	4576
Total non-grey	4050	2644
Total genes	7220	7220

Figure 4 and 5 shows the gene clustering dendrograms with module colour bars for both networks (power 14 signed, power 7 unsigned). The signed dendrogram shows visibly deeper, more sharply resolved branch structure feeding several distinct, large colour blocks (turquoise, blue, green/red/yellow cluster, plus additional modules on the right), consistent with the larger, more evenly sized module set in Table 3.

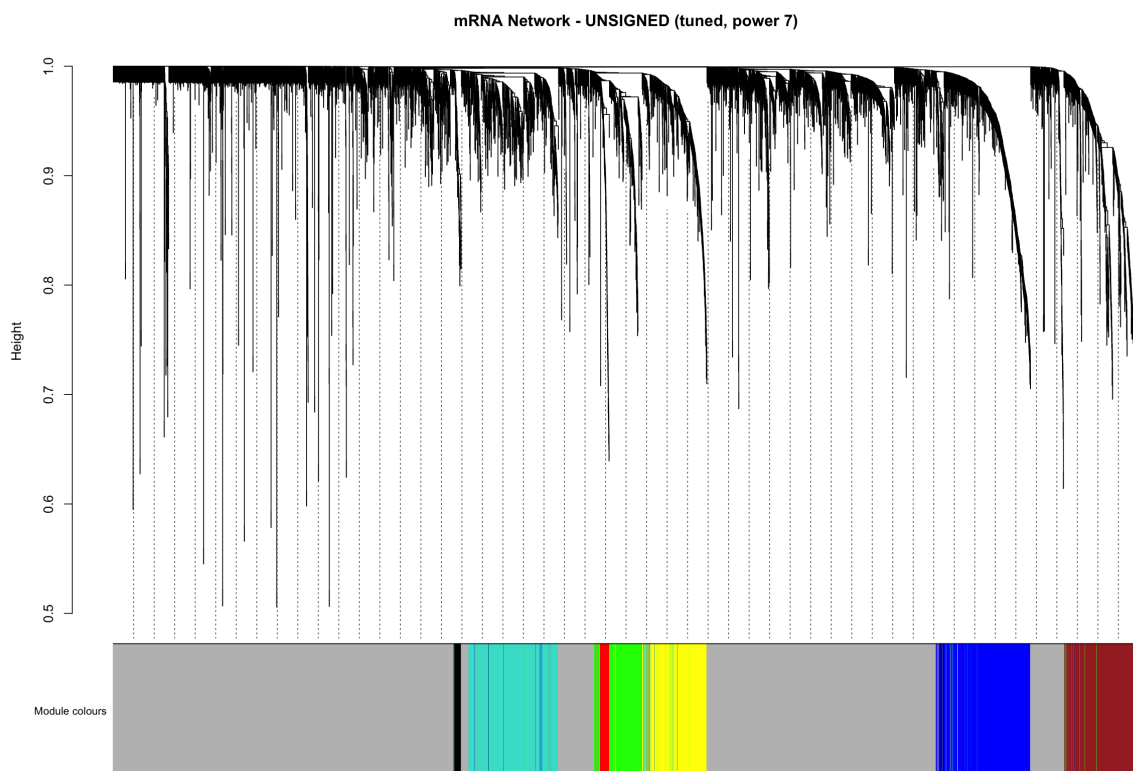


Figure 4: Gene clustering dendrogram with module colour assignment for the tuned unsigned mRNA network.

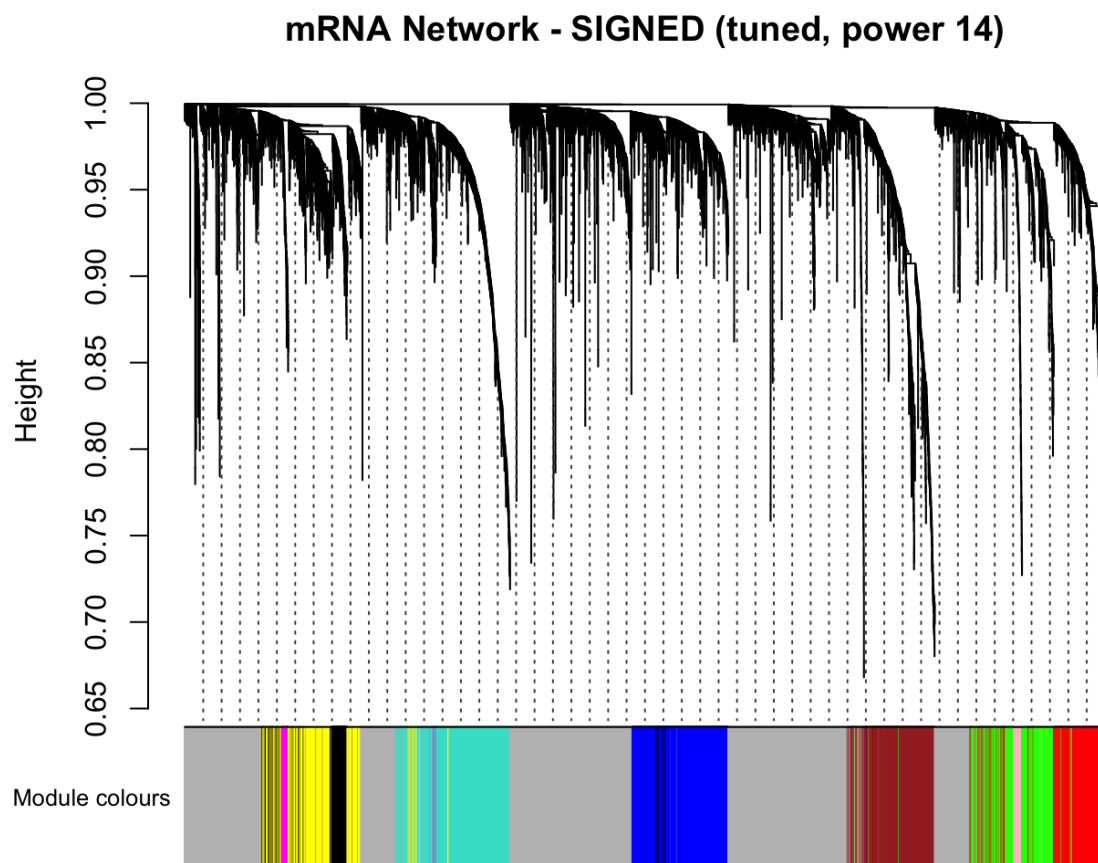


Figure 5: Gene clustering dendrogram with module colour assignment for the tuned signed mRNA network.

2.6 Module Eigengene Correlation Structure

Figure 6 and 7 shows the pairwise correlation of module eigengenes within both the networks.

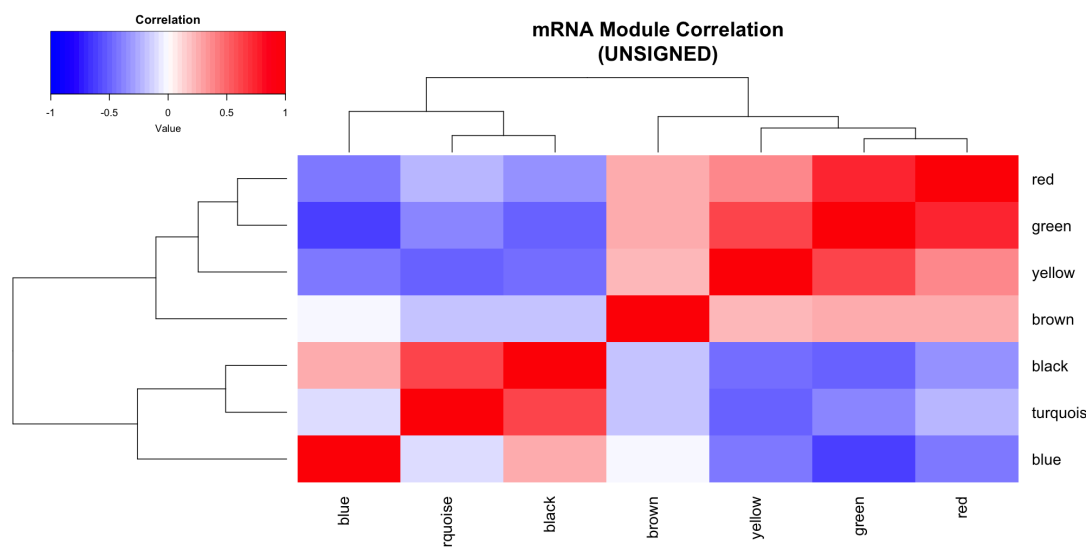


Figure 6: Module eigengene correlation heatmap for the unsigned mRNA network, with hierarchical clustering of modules by eigengene correlation.

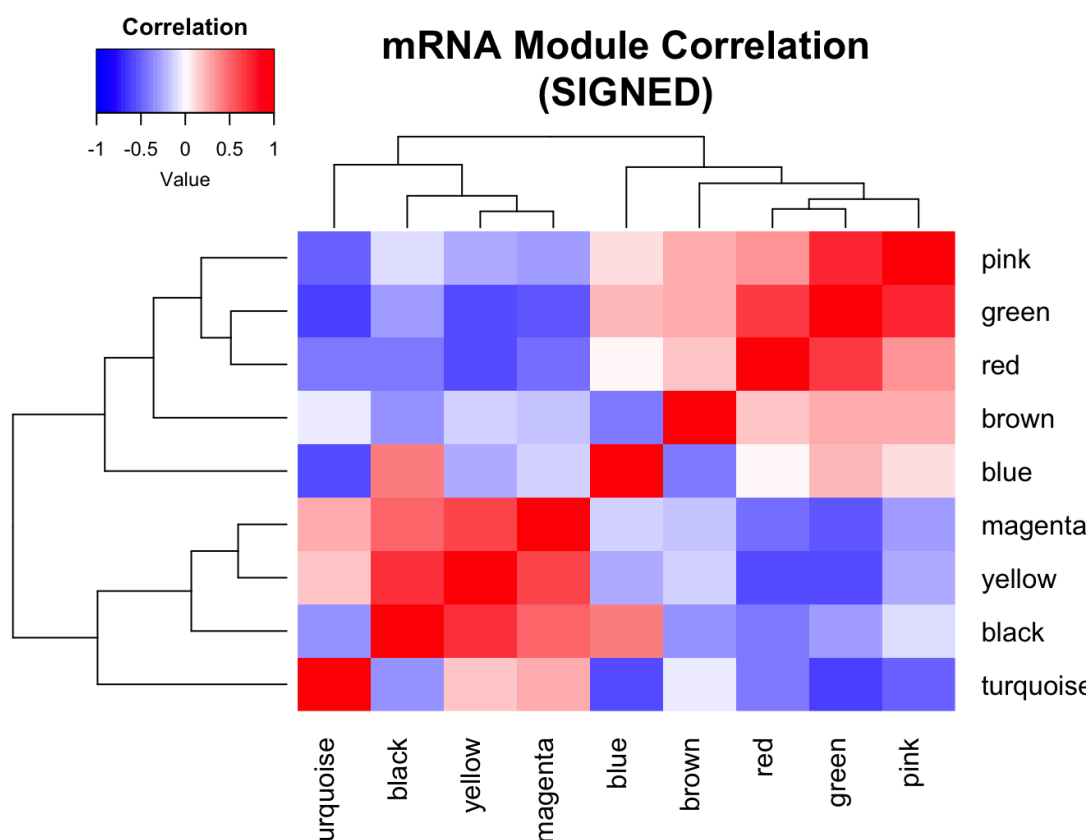


Figure 7: Module eigengene correlation heatmap for the signed mRNA network, with hierarchical clustering of modules by eigengene correlation.

2.7 Module–Trait Relationships

Figures 8 and 9 show the full module–subtype correlation heatmaps (Pearson r and Student p -value per cell) for the signed and unsigned networks respectively; the underlying values for the signed network are reproduced in Table 4.

Table 4: Signed module–subtype Pearson correlations (all p -values $< 10^{-4}$ except where noted; *n.s. is not significant*).

Module	Basal	HER2	LumA	LumB
turquoise	0.56	0.14	−0.73	0.29
blue	−0.86	−0.16	0.62	0.15
green	−0.26	−0.09 (n.s.)	0.49	−0.30
black	−0.36	−0.01 (n.s.)	0.24	0.06 (n.s.)
yellow	0.23	0.08	−0.24	0.02 (n.s.)
brown	0.23	0.11	−0.13	−0.13
red	−0.16	0.01 (n.s.)	0.27	−0.19
pink	−0.14	−0.11	0.37	−0.25
magenta	0.14	−0.03 (n.s.)	−0.26	0.20

The dominant pattern across both heatmaps is a strong ***Basal–LumA opposition***: turquoise and blue are the two largest-magnitude modules (blue: $r = -0.86$ with Basal, $r = 0.62$ with LumA; turquoise: $r = 0.56$ with Basal, $r = -0.73$ with LumA), and every other module’s correlation sign is consistent with belonging to one side or the other of this same axis. ***HER2 is comparatively weakly represented across all modules*** (largest $|r| = 0.16$), and several module–HER2 cells are not significant ($p > 0.05$), which matches the smaller HER2 sample size ($n=75$) and its known intermediate transcriptional profile relative to Basal and LumA. The blue module’s known top hub gene, *ESR1* (Section 2.6/hub-gene table; $kME = 0.883$), is the canonical estrogen-receptor gene whose strong negative correlation with Basal and positive correlation with LumA is the expected direction (LumA tumours are ER-positive, Basal tumours are predominantly ER-negative).

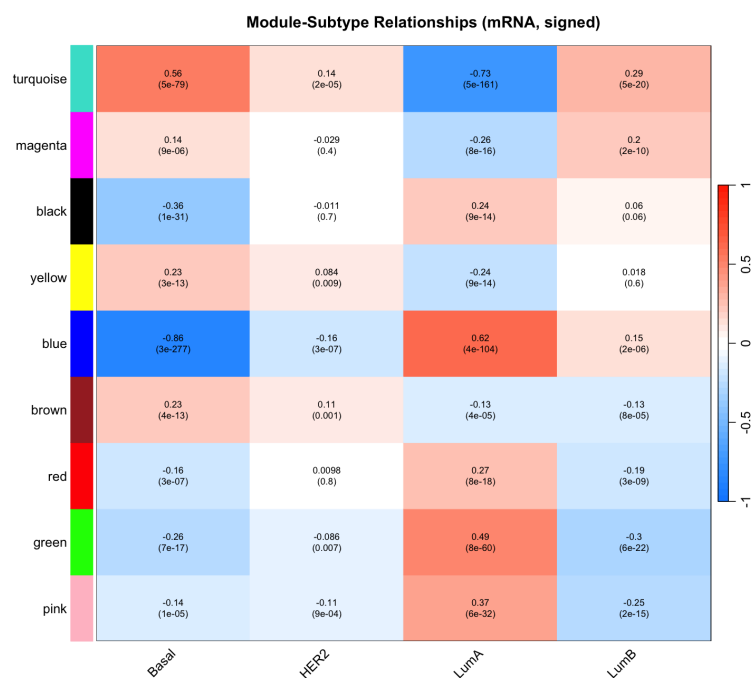


Figure 8: Module-subtype relationships, signed mRNA network. Each cell reports Pearson r (top) and p -value (bottom).

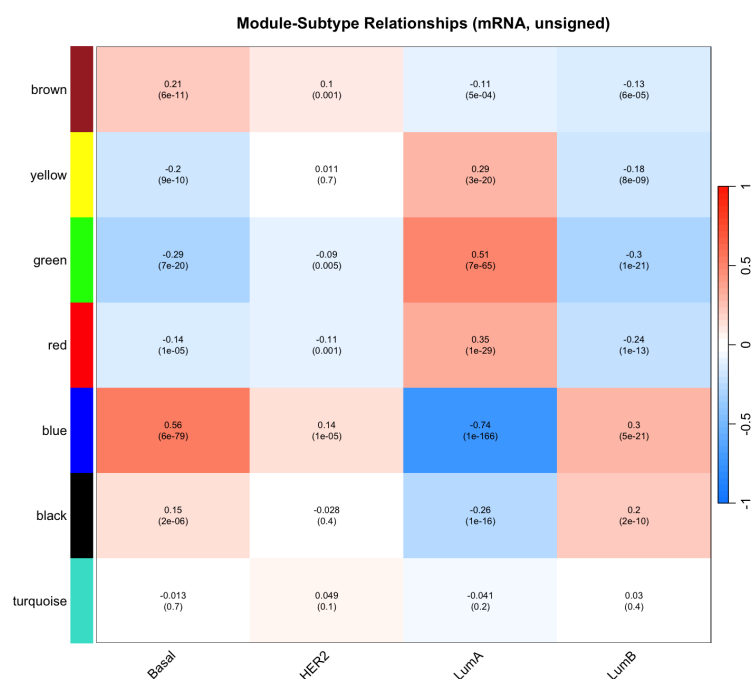


Figure 9: Module-subtype relationships, unsigned mRNA network. Each cell reports Pearson r (top) and p -value (bottom).

2.8 Hub Genes

The kME-ranked top hub gene per signed module was: turquoise – *MCM10* (kME 0.914); blue – *ESR1* (kME 0.883); green – *ERG* (kME 0.901); black – *MCRIP2* (kME 0.895); yellow – *NAA10* (kME 0.856); brown – *SNX20* (kME 0.959); red – *GLT8D2* (kME 0.932); pink – *CIDEA* (kME 0.937); magenta – *PUF60* (kME 0.921). ***No gene appeared as a member of more than one module in either network (confirmed).*** Notably, the same underlying genes surface as top hubs of *different-coloured* modules in the unsigned network (e.g. *MCM10* tops signed-turquoise but unsigned-blue; *PUF60* tops signed-magenta but unsigned-black; *CIDEA* tops signed-pink but unsigned-red). Module colour identities are arbitrary and are not comparable across the two runs, only the underlying gene composition is.

3 Reasons for Choosing Signed Network

1. **Prevention of Signal Cancellation:** Unsigned adjacency uses absolute correlation values ($|r|$), grouping strongly positively and strongly negatively correlated genes into the same module. In this dataset, the 639-gene unsigned **turquoise** module shows near-zero correlation across all PAM50 subtypes ($|r| \leq 0.049$, $p > 0.1$). When constructed signed, these same genes separate into the **blue** and **turquoise** modules, yielding strong subtype correlations ($|r|$ up to 0.86). Computing eigengenes from anti-correlated genes in the unsigned network causes expression signals to cancel out.
2. **Higher Coverage:** The signed network assigned 56.1% (4,050/7,220) of genes to non-grey modules, compared to 36.6% (2,644/7,220) in the unsigned network under identical parameters (`minModuleSize` and `mergeCutHeight`). Consequently, the unassigned (**grey**) pool was reduced from 4,576 genes to 3,170 genes.
3. **Directional Biological Interpretation:** Signed networks preserve the direction of gene co-expression, allowing module correlations to directly indicate up- or down-regulation relative to PAM50 subtypes and key biomarkers (e.g., *ESR1*). Unsigned modules mix oppositely regulated genes, preventing unambiguous directional interpretation.

4 GO Enrichment Analysis of the Signed Network

GO over-representation analysis (`clusterProfiler::enrichGO()`, ontologies BP/CC/MF, background = all 7220 genes entering WGCNA, BH-adjusted $p < 0.05$ and $q < 0.05$) was run for all nine non-grey signed modules. Every module returned significant terms in at least one ontology. The combined BP/CC/MF dot plots are shown below (top 10 terms per ontology, ranked by gene ratio), one subsection per module, together with a short interpretation that cross-references each module's dominant subtype correlation (Section 2.7, Table 4) and its top kME hub gene (Section 2.8).

4.1 Turquoise: Cell Cycle / DNA Replication

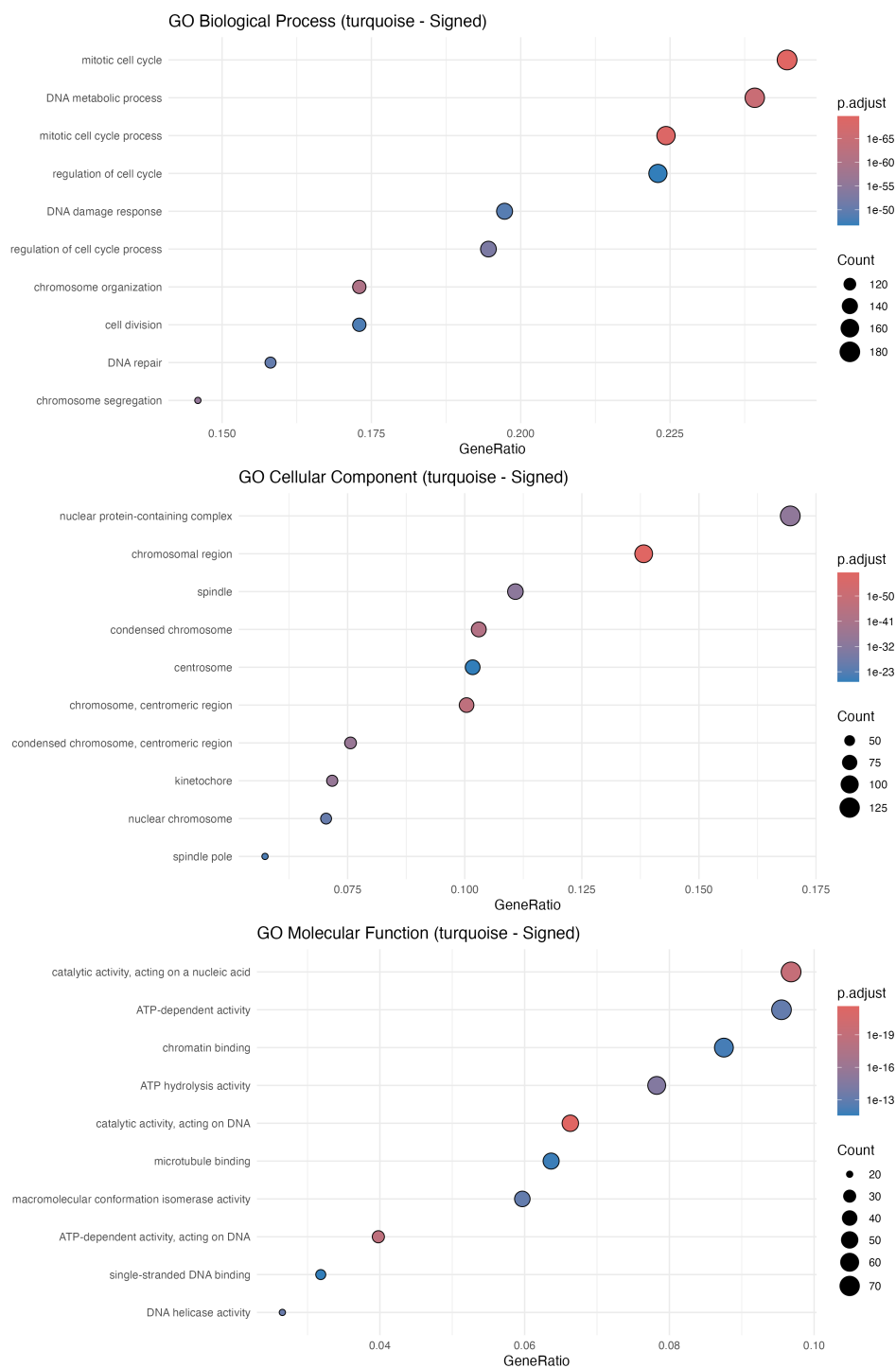


Figure 10: GO enrichment, signed turquoise module (785 genes).

turquoise is a **proliferation / cell-cycle** module: every top BP term is cell-cycle or DNA-metabolism related (mitotic cell cycle, DNA metabolic process, DNA damage response, chromosome segregation), CC terms are dominated by mitotic machinery (spindle, kinetochore, condensed chromosome, centrosome), and MF terms are dominated by DNA-acting catalytic/ATP-dependent activities (DNA helicase activity, catalytic activity on DNA/nucleic acid). This is fully consistent with turquoise's strong *positive* correlation with Basal ($r = 0.56$) and strong

negative correlation with LumA ($r = -0.73$): Basal tumours a highly proliferative PAM50 sub-type, LumA the least. The module's top hub gene, *MCM10* (kME = 0.914), is a core component of the DNA replication machinery.

4.2 Blue: Ciliated/Differentiated Epithelial Identity

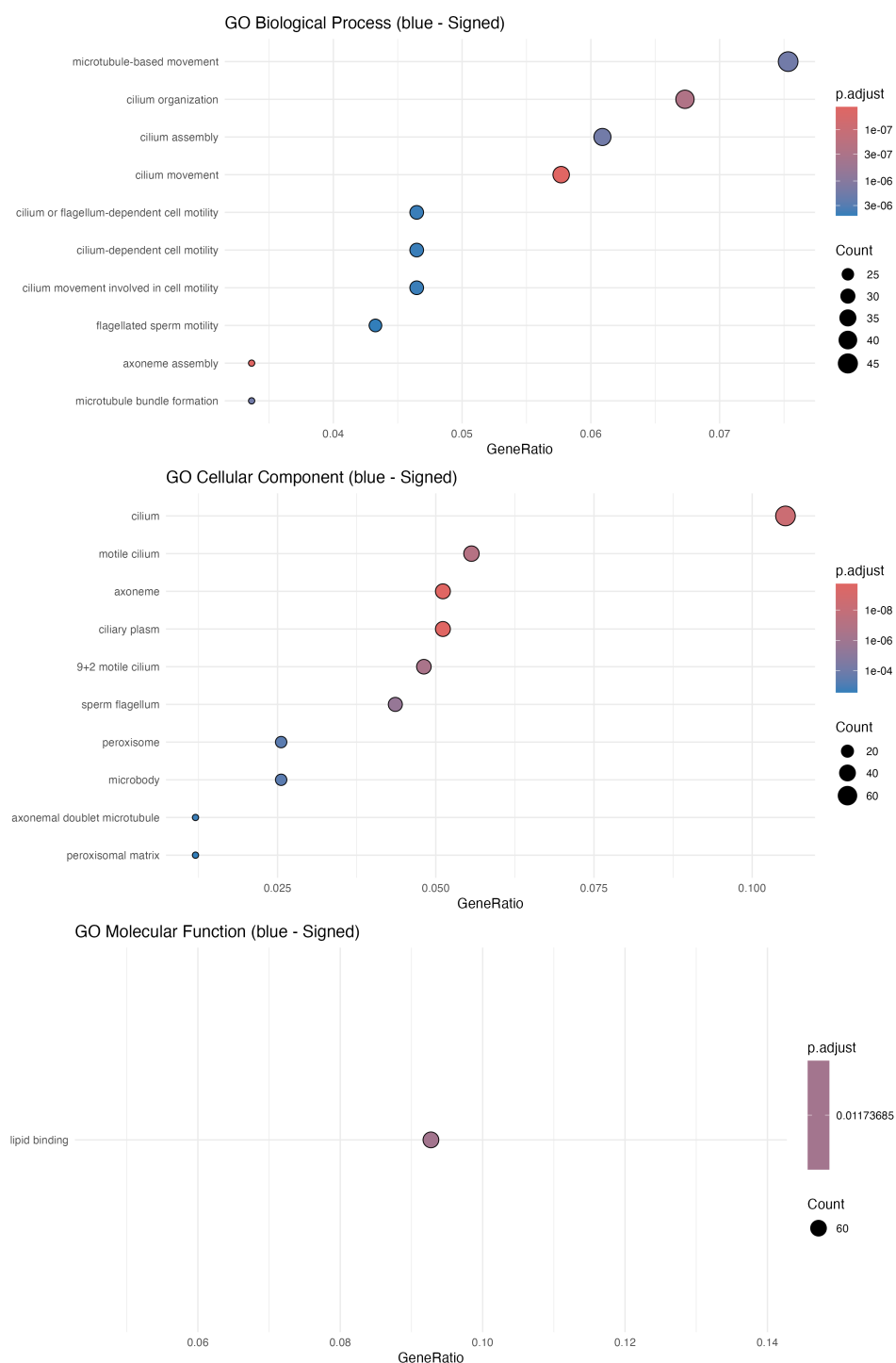


Figure 11: GO enrichment, signed blue module (715 genes).

Cilium organisation/assembly/movement in BP, cilium/axoneme/motile cilium/9+2 motile cilium in CC, plus a peroxisome/lipid-binding signature. This is the strongest trait-correlated

module in the whole analysis (Basal $r = -0.86$, LumA $r = 0.62$), and its top hub gene is *ESR1* (kME = 0.883), the estrogen receptor gene that molecularly defines the luminal subtypes.

4.3 Green: Vasculature / Endothelium

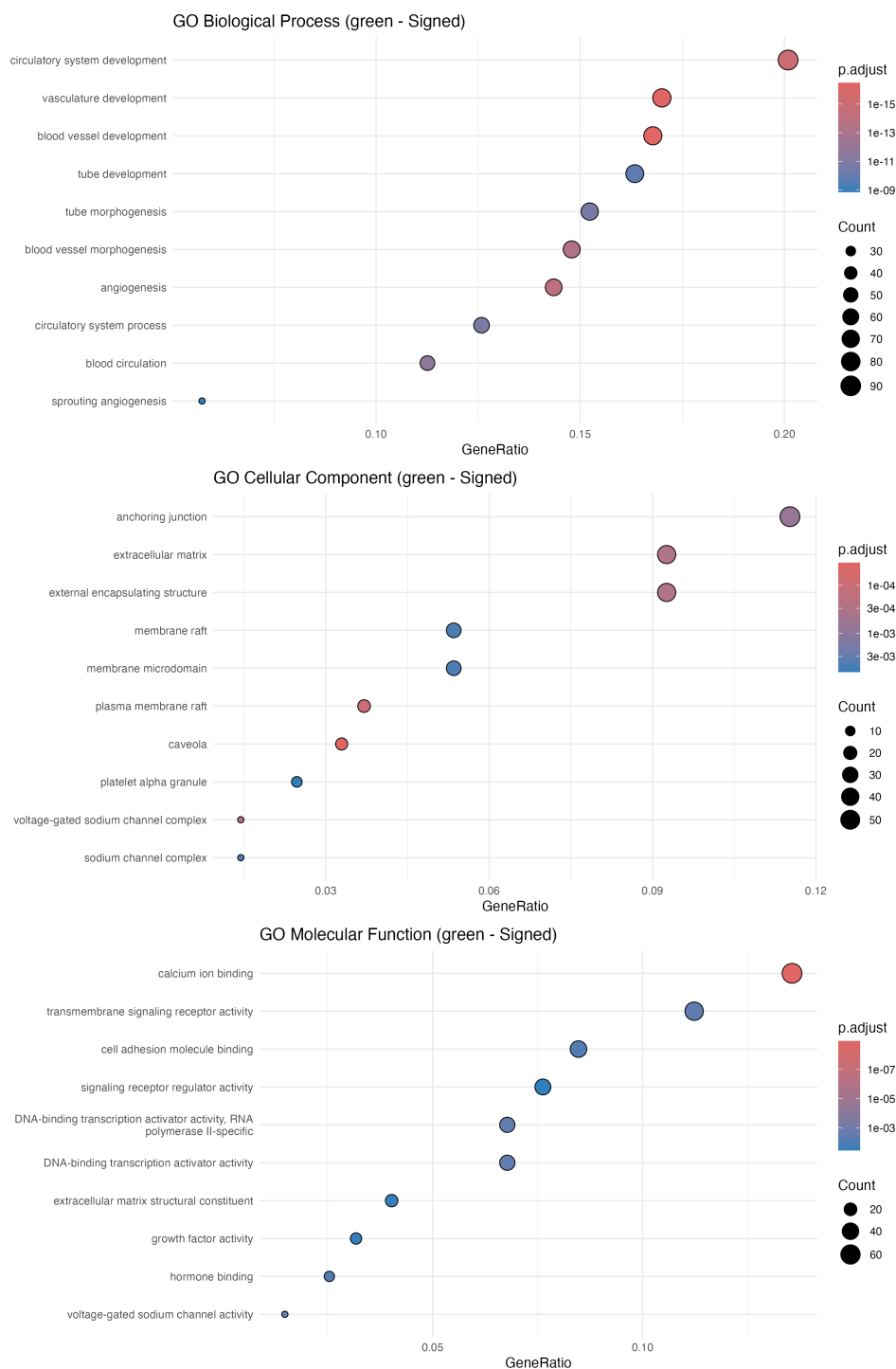


Figure 12: GO enrichment, signed green module (503 genes).

green's BP terms are dominated by **blood-vessel development and angiogenesis** (circulatory system development, vasculature/blood vessel development and morphogenesis, angiogenesis), with CC terms pointing to junctional/matrix structures (anchoring junction, extracellular

matrix) and MF terms centred on calcium-ion binding and transmembrane signalling receptor activity. This tracks green's positive correlation with LumA ($r = 0.49$) and negative correlation with Basal/LumB, and is corroborated strongly by its top hub gene *ERG* ($kME = 0.901$), a transcription factor that is one of the most specific markers of vascular endothelial cells.

4.4 Black: Mitochondrial / Organelle Envelope

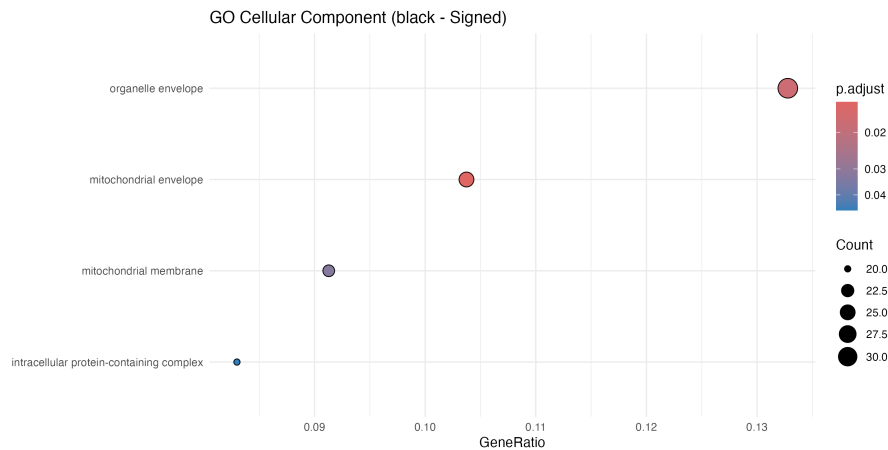


Figure 13: GO enrichment, signed **black** module (262 genes). Only the CC ontology returned significant terms.

black returned significant terms only in Cellular Component, all pointing to **mitochondrial/organelle membranes** (organelle envelope, mitochondrial envelope, mitochondrial membrane, intracellular protein-containing complex). This is a comparatively generic signature. It correlates negatively with Basal ($r = -0.36$) and positively with LumA ($r = 0.24$), suggesting a broadly higher mitochondrial-membrane/organelle content in the less proliferative LumA tumours, **though the module's top hub gene *MCRIP2* ($kME = 0.895$) is not itself a canonical mitochondrial gene, and the absence of any significant BP or MF term means this module's biology is less sharply defined than turquoise, blue, or green.**

4.5 Yellow: Ribosome Biogenesis / Oxidative Phosphorylation

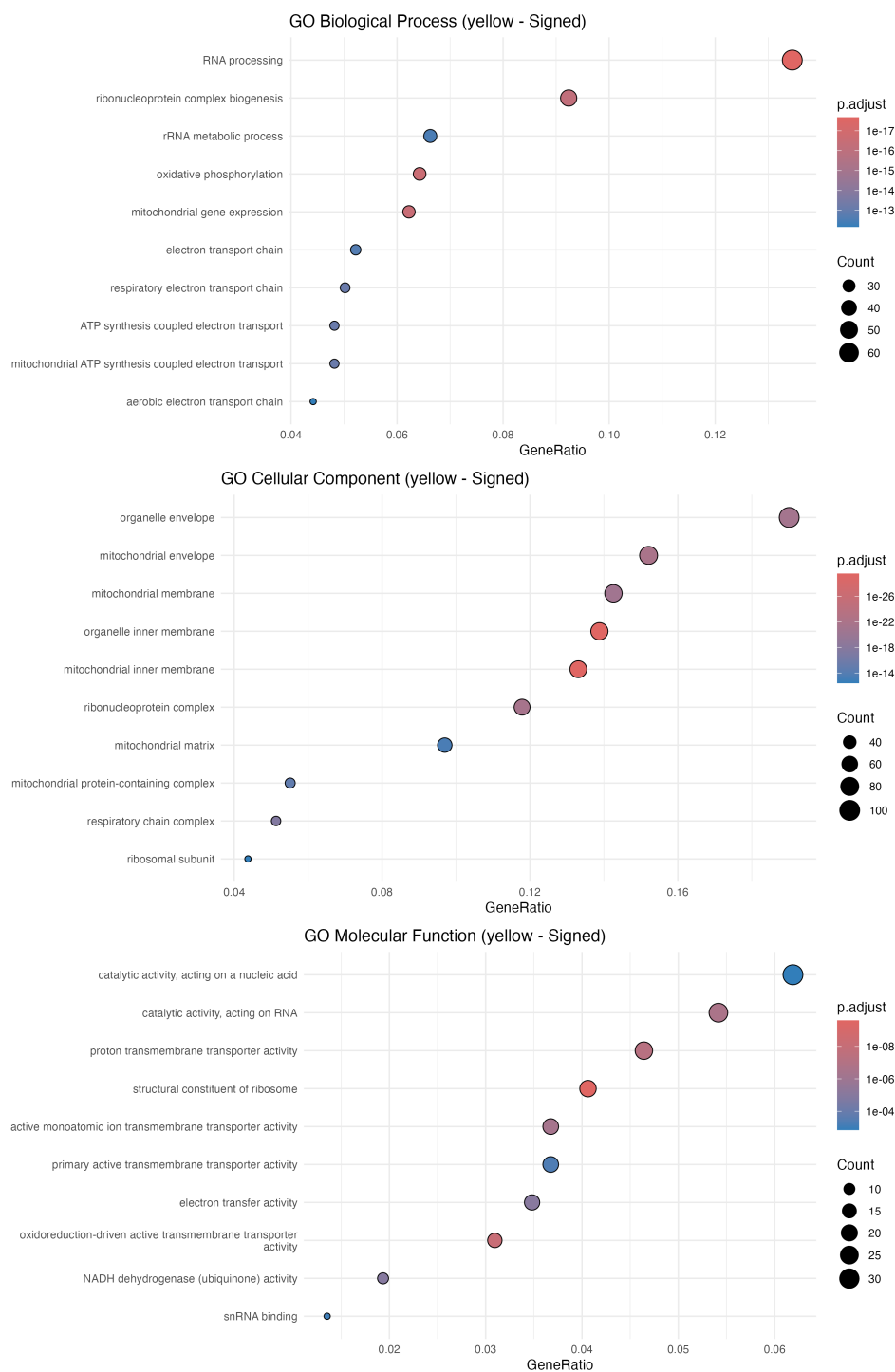


Figure 14: GO enrichment, signed **yellow** module (544 genes).

yellow shows a mixed but coherent **translational/metabolic** signature: RNA processing and ribonucleoprotein complex biogenesis dominate BP alongside oxidative phosphorylation and mitochondrial gene expression; CC is almost entirely mitochondrial (organelle/mitochondrial envelope, mitochondrial inner membrane, mitochondrial matrix) plus ribonucleoprotein complex; MF is split between nucleic-acid catalytic activity/structural constituent of ribosome and transmembrane transporter/electron-transfer activities (consistent with the electron transport

chain machinery). Yellow correlates positively with Basal ($r = 0.23$) and negatively with LumA ($r = -0.24$). The more proliferative Basal tumours upregulate both ribosome biogenesis (to support high protein synthesis demand) and oxidative-phosphorylation/mitochondrial gene expression programmes together. The top hub gene, *NAA10* (kME = 0.856), is an N-terminal acetyltransferase implicated in regulating protein synthesis and stability.

4.6 Brown: Immune / Leukocyte Activation

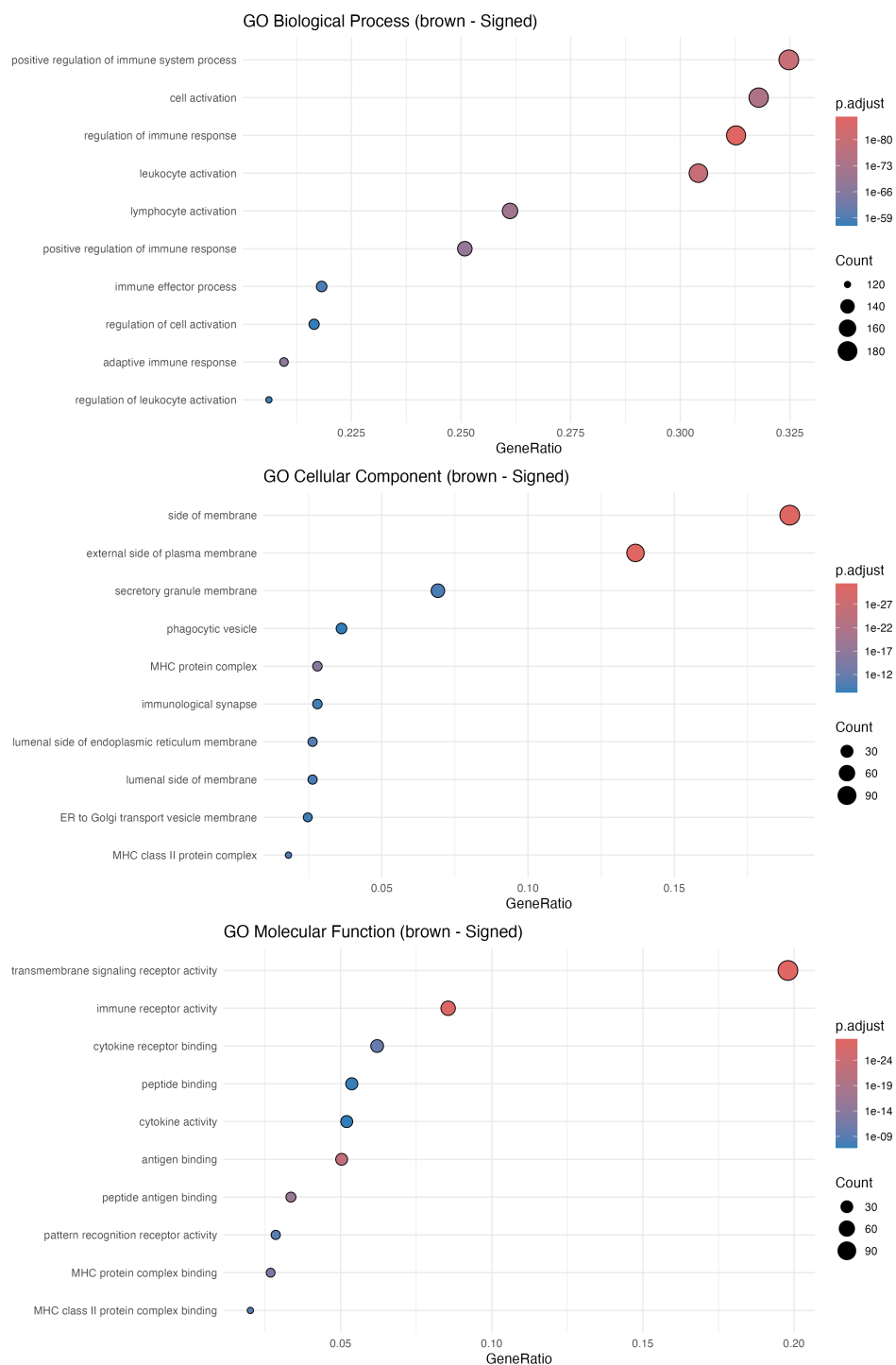


Figure 15: GO enrichment, signed brown module (619 genes).

brown is a clear **immune/inflammatory** module: BP is dominated by immune system process regulation, cell/leukocyte/lymphocyte activation and adaptive immune response; CC includes MHC protein complex, immunological synapse, and phagocytic vesicle; MF is dominated by immune receptor activity, cytokine receptor binding/activity, and antigen/peptide-antigen binding, including MHC class II binding specifically. This module shows only a weak positive correlation with Basal ($r = 0.23$) and weak negative correlations with LumA/LumB. The module's top hub gene is *SNX20* (kME = 0.959), a sorting nexin expressed predominantly in immune cells.

4.7 Red: Extracellular Matrix / Collagen

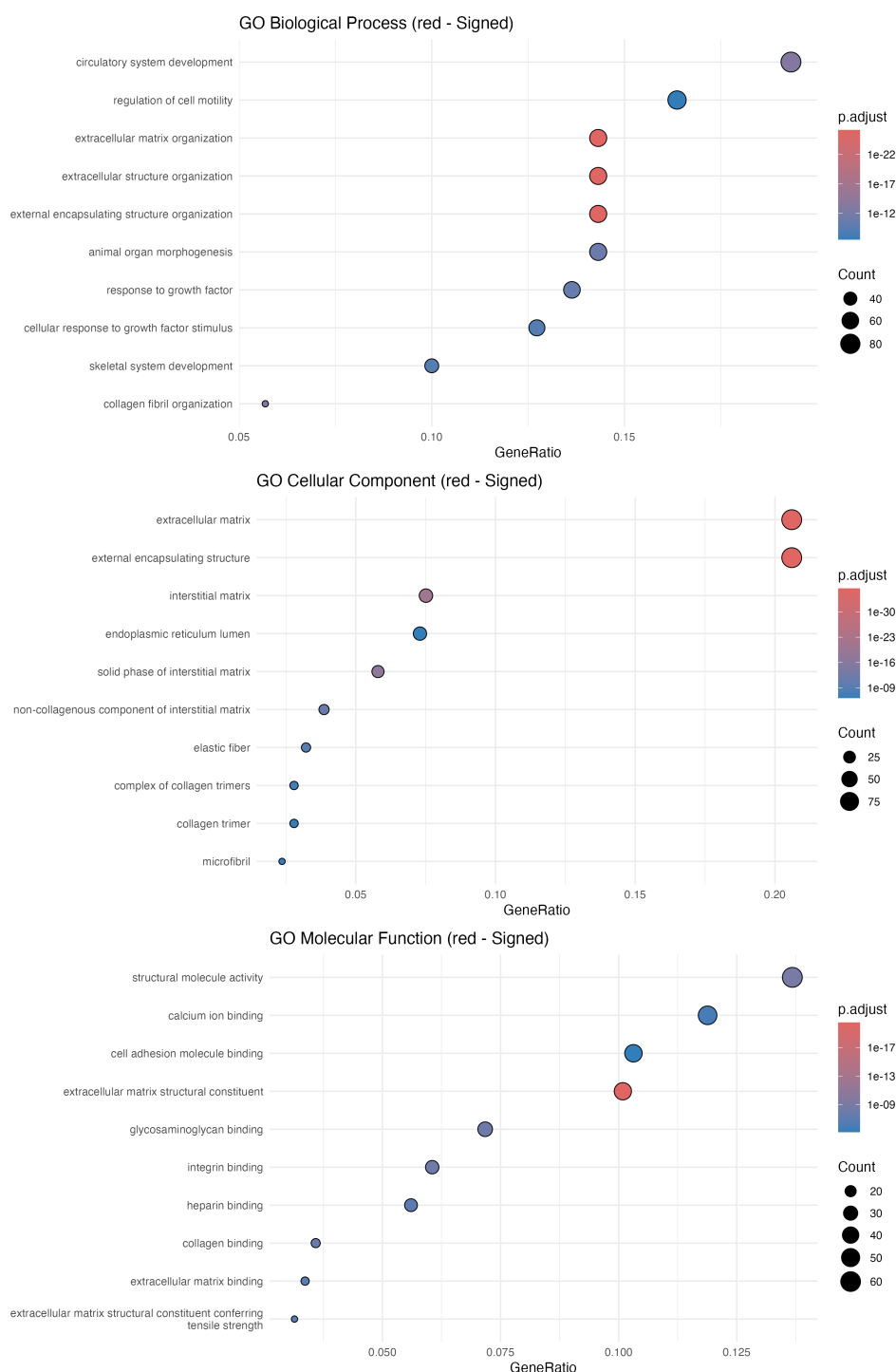


Figure 16: GO enrichment, signed red module (472 genes).

red is an **extracellular matrix / stromal** module, closely related to green but more specifically collagen-centred: BP terms include extracellular matrix/structure organisation and collagen fibril organisation; CC is dominated by extracellular matrix, interstitial matrix, collagen trimer, and elastic fibre; MF is dominated by structural molecule activity, collagen binding, integrin binding, and glycosaminoglycan binding. Red correlates positively with LumA ($r = 0.27$) and negatively with Basal ($r = -0.16$), again pointing to differences in stromal/matrix

composition across subtypes. The top hub gene *GLT8D2* ($kME = 0.932$) is a glycosyltransferase implicated in collagen glycosylation, which lines up precisely with the collagen-heavy enrichment result.

4.8 Pink: Lipid Storage / Adipocyte Signature

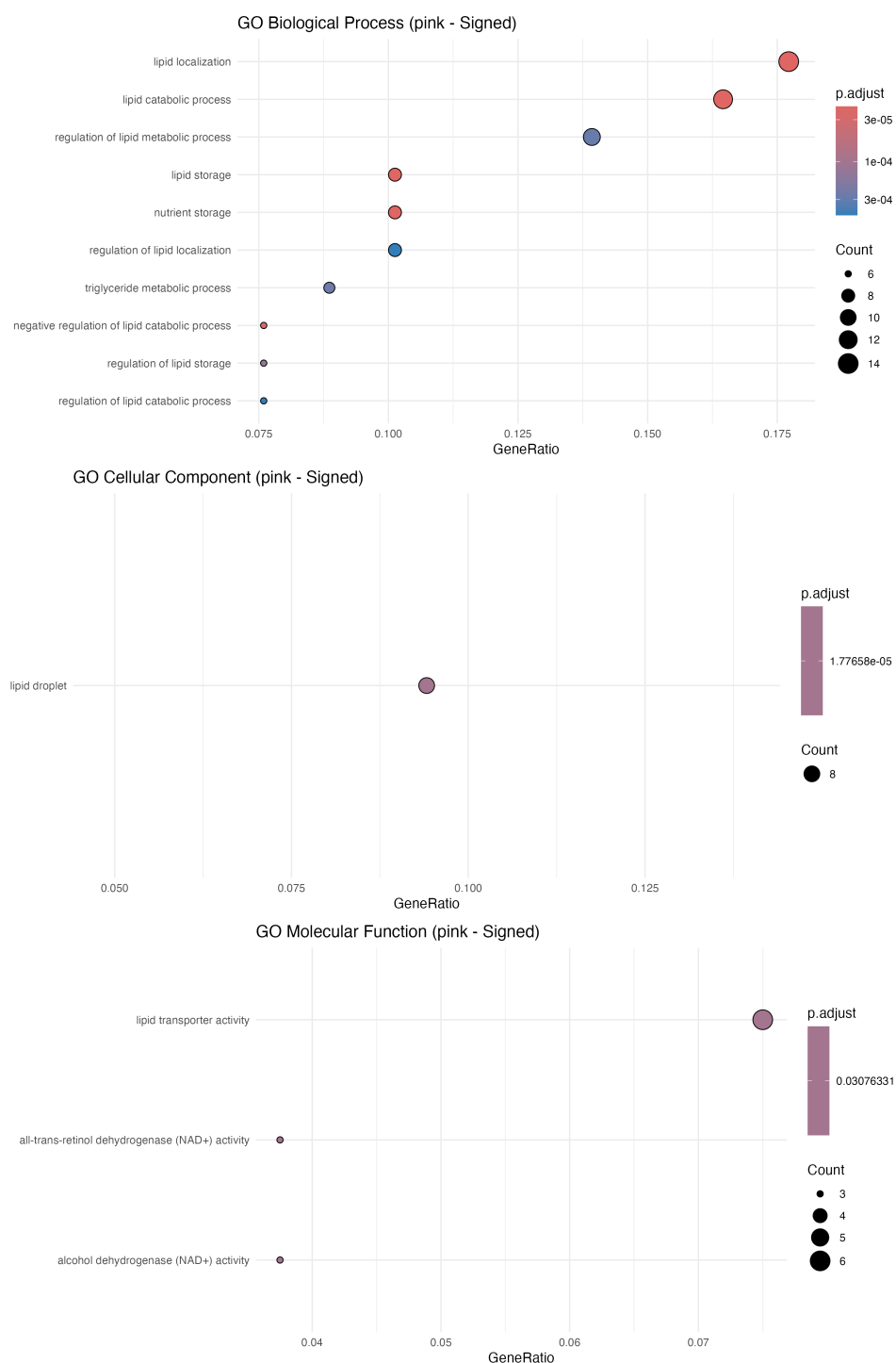


Figure 17: GO enrichment, signed **pink** module (85 genes).

pink is a **lipid metabolism / adipocyte** module: BP is dominated by lipid localisation, catabolism, storage and nutrient storage; the sole significant CC term is lipid droplet; MF

includes lipid transporter activity and NAD⁺-dependent dehydrogenase activities. Pink correlates positively with LumA ($r = 0.37$) and negatively with Basal/LumB, and its top hub gene is *CIDEA* (kME = 0.937), a canonical adipocyte lipid-droplet gene. This strongly suggests pink is capturing variation in adjacent adipose-tissue content within the tumour samples (mammary tumours are embedded in adipose stroma, and the amount of contaminating/adjacent fat tissue captured on a section can vary), which happens to differ systematically by subtype rather than reflecting an intrinsic cancer-cell lipid programme.

4.9 Magenta: Vesicle Trafficking

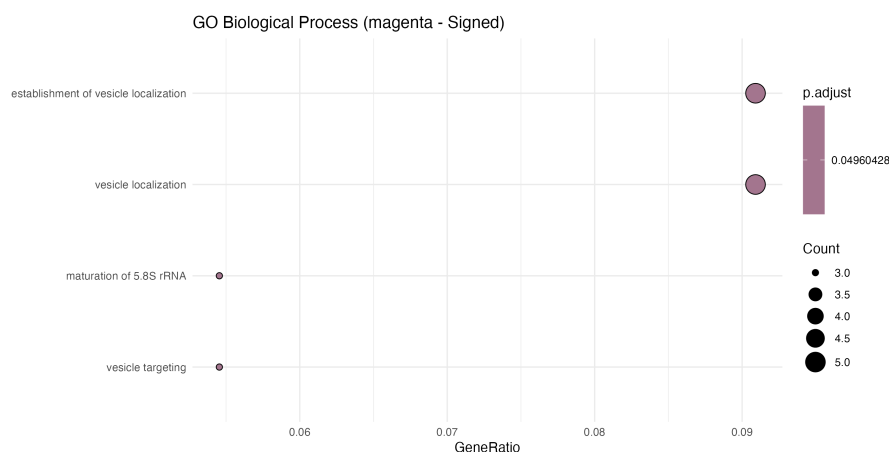


Figure 18: GO enrichment, signed **magenta** module (65 genes). Only the BP ontology returned significant terms.

magenta is the smallest signed module (65 genes) and returned significant terms only in Biological Process, all related to **vesicle localisation/targeting** (establishment of vesicle localization, vesicle localization, vesicle targeting) plus one unrelated term (maturation of 5.8S rRNA), with very small gene counts per term (3–5 genes). The top hub gene is *PUF60* (kME = 0.921), a pre-mRNA splicing factor.

4.10 Summary

Table 5: One-line functional summary per signed module, its dominant PAM50 association, and its top hub gene.

Module	Dominant biology	Strongest subtype link	Top hub (kME)
turquoise	Cell cycle / DNA replication	Basal ↑ / LumA ↓	MCM10 (0.914)
blue	Ciliated / luminal differentiation	Basal ↓ / LumA ↑	ESR1 (0.883)
green	Vasculature / endothelium	LumA ↑	ERG (0.901)
black	Mitochondrial / organelle envelope	LumA ↑ / Basal ↓	MCRIIP2 (0.895)
yellow	Ribosome biogenesis / OXPHOS	Basal ↑ / LumA ↓	NAA10 (0.856)
brown	Immune / leukocyte activation	Basal ↑ (weak)	SNX20 (0.959)
red	Extracellular matrix / collagen	LumA ↑ / Basal ↓	GLT8D2 (0.932)
pink	Lipid storage / adipocyte	LumA ↑ / Basal,LumB ↓	CIDEA (0.937)
magenta	Vesicle trafficking (weak/small)	Basal ↑ (weak)	PUF60 (0.921)